rfixest

Fast fixed-effects regressions in Stata, powered by R's `fixest`

A practical guide • version 1.1.0 • https://github.com/blacaber8/rfixest

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1 What rfixest does

`rfixest` estimates regressions with many high-dimensional fixed effects (individuals, firms, years and their interactions) from inside Stata. It sends the data to the R package `fixest`, one of the fastest implementations of these estimators, and brings the results back to Stata. You type a Stata command and get a Stata results table, and R works in the background without opening any window.

- The syntax is the same as `reghdfe`, so existing code needs only the command name changed.
- Point estimates and standard errors match `reghdfe` (OLS) and `ppmlhdfe` (Poisson).

- Several outcomes can be estimated in one call, with the fixed effects computed once.
- The results are a regular Stata estimation, so `esttab`, `estimates store`, `test`, `lincom` and `e(sample)` work as usual.

2 Installation

Requirements. Stata 16 or later (`rfixest` uses Stata data frames, introduced in version 16) and R 4.0 or later. It works on Windows, macOS and Linux.

Step 1. Install R. Download it from https://cran.r-project.org and install it with the default options. Any recent version works. You do not need to open R afterwards.

Step 2. Install rfixest in Stata. Type in Stata:

```
net install rfixest, from("https://raw.githubusercontent.com/blacaber8/rfixest/main/") replace
```

Step 3. Run the setup once.

```
rfixest setup
```

The setup finds R on your computer, installs the R package `fixest` if it is missing (this takes a minute or two the first time), and remembers where R is for all future Stata sessions. When it finishes it prints a line like

```
rfixest is ready. R: C:/Program Files/R/R-4.3.3/bin/x64/Rscript.exe; fixest 0.12.0.
```

If R is installed in an unusual folder, give the location of `Rscript`:

```
rfixest setup, rscript("D:/Programs/R/R-4.4.1/bin/x64/Rscript.exe")
```

3 Syntax

```
rfixest depvar [indepvars] [if] [in] [weight] [, options]
rfixest (depvar1 depvar2 ...) [indepvars] [if] [in] [weight] [, options]
rfixest setup [, rscript(path)]
```

indepvars may contain factor variables (`i.group`, `c.x#i.group`) and time-series operators. Weights may be `aweight` or `pweight`.

Options

Model	
`absorb(`*terms*`)`	fixed effects, for example `absorb(id year)` or `absorb(id industry#year)`. Without `absorb()` the model has a constant and no fixed effects. Writing `absorb(FE_id=id year)` also saves the fixed effect of `id` in the new variable `FE_id`.
`family(ols|poisson|negbin)`	type of model; default `ols`
`keepsingletons`	keep groups with a single observation (dropped by default, as in `reghdfe`)
Standard errors	
`vce(robust)`	heteroskedasticity-robust
`vce(cluster` *v1* `[`*v2*`])`	clustered; two or more variables give multiway clustering
`cluster(`*varlist*`)`	same as `vce(cluster` *varlist*`)`
`dfresid`	t tests with $N-K$ degrees of freedom instead of $G-1$ when clustering
`ppmlse`	no small-sample adjustment of the variance, as in `ppmlhdfe`
Reporting and storing	
`residuals(`*newvar*`)`	save the residuals in a new variable
`estore(`*name*`)`	store the estimates; with several outcomes they are stored as *name*1, *name*2, ...
`estart(#)`	first number used for the stored names; default 1
`quietly`	do not show the table
`level(#)`	confidence level; default 95
`fullsample`	`e(sample)` also marks dropped singletons
`timeit`	show the time spent in Stata and in R
Technical	
`threads(#)`	threads used by `fixest`; default is `fixest`'s own choice
`tol(#)`	convergence tolerance of the fixed effects; default 10^{-6}
`rscript(`*path*`)`	location of `Rscript` for this call only
`tmpdir(`*path*`)`	folder for temporary files; default is Stata's temporary folder
`keepfiles`	keep the temporary files, for debugging

4 Examples

All examples use the NLS Women panel that comes with Stata (`webuse nlswork`), with about 28,000 observations of 4,700 women between 1968 and 1988. The outputs below are exactly what Stata prints.

4.1 Individual and year fixed effects

The basic model, with standard errors clustered by individual.

```
. rfixest ln_wage tenure ttl_exp, absorb(idcode year) vce(cluster idcode)

Linear regression with fixed effects (fixest)     Number of obs    =     27,549
Dependent variable: ln_wage                       Residual df      =      4,146
Fixed effects: idcode year                        R-squared        =     0.6744
Std. errors: cluster (idcode)                     Adj R-squared    =     0.6164
                                                  Within R-sq.     =     0.0477
                                                  Root MSE         =     0.2947
                                                  Clusters         =      4,147
------------------------------------------------------------------------------
     ln_wage |    Coef.     Std. err.    t     P>|t|    [95% conf. interval]
-------------+----------------------------------------------------------------
      tenure |  .0106811    .0014932    7.15   0.000     .0077537   .0136086 ***
     ttl_exp |   .032636    .0024388   13.38   0.000     .0278546   .0374174 ***
------------------------------------------------------------------------------
* p<0.10, ** p<0.05, *** p<0.01
Singletons or perfectly predicted observations dropped: 552
Degrees of freedom absorbed by fixed effects: 4,161
```

4.2 The same model with reghdfe

Coefficients, standard errors, R^2 and number of observations are identical.

```
. reghdfe ln_wage tenure ttl_exp, absorb(idcode year) vce(cluster idcode)
(dropped 552 singleton observations)
(MWFE estimator converged in 8 iterations)

HDFE Linear regression                            Number of obs   =     27,549
Absorbing 2 HDFE groups                           F(   2,   4146) =     266.45
Statistics robust to heteroskedasticity           Prob > F        =     0.0000
                                                  R-squared       =     0.6744
                                                  Adj R-squared   =     0.6164
                                                  Within R-sq.    =     0.0477
Number of clusters (idcode)  =      4,147         Root MSE        =     0.2947

                            (Std. err. adjusted for 4,147 clusters in idcode)
------------------------------------------------------------------------------
             |               Robust
     ln_wage | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
      tenure |   .0106811   .0014932     7.15   0.000     .0077537    .0136086
     ttl_exp |    .032636   .0024388    13.38   0.000     .0278546    .0374174
       _cons |   1.440671   .0130842   110.11   0.000     1.415019    1.466323
------------------------------------------------------------------------------

Absorbed degrees of freedom:
-----------------------------------------------------+
 Absorbed FE | Categories  - Redundant  = Num. Coefs |
-------------+---------------------------------------|
      idcode |      4147        4147           0    *|
```

```
        year |        15           1          14     |
-------------------------------------------------------+
* = FE nested within cluster; treated as redundant for DoF computation
```

4.3 Interacted fixed effects and factor variables

`ind_code#year` absorbs a separate fixed effect for each industry and year. Factor variables work as in any Stata command.

```
. rfixest ln_wage tenure ttl_exp i.union, absorb(idcode ind_code#year) cluster(idcode)

Linear regression with fixed effects (fixest)     Number of obs    =     18,252
Dependent variable: ln_wage                       Residual df      =      3,450
Fixed effects: idcode ind_code#year               R-squared        =     0.7699
Std. errors: cluster (idcode)                     Adj R-squared    =     0.7135
                                                  Within R-sq.     =     0.0633
                                                  Root MSE         =     0.2481
                                                  Clusters         =      3,451
------------------------------------------------------------------------------
     ln_wage |    Coef.     Std. err.    t     P>|t|    [95% conf. interval]
-------------+----------------------------------------------------------------
      tenure |  .0088179     .001276    6.91   0.000      .006316   .0113197 ***
     ttl_exp |  .0346327    .0027624   12.54   0.000     .0292167   .0400487 ***
    0b.union |    (base)
     1.union |  .0896231     .008876   10.10   0.000     .0722204   .1070257 ***
------------------------------------------------------------------------------
* p<0.10, ** p<0.05, *** p<0.01
Singletons or perfectly predicted observations dropped: 673
Degrees of freedom absorbed by fixed effects: 3,590
```

4.4 Two-way clustering

Listing two variables in `cluster()` clusters by both. As in `reghdfe`, the t tests use the number of clusters of the smaller dimension minus one.

```
. rfixest ln_wage tenure ttl_exp, absorb(idcode year) cluster(idcode ind_code)

Linear regression with fixed effects (fixest)     Number of obs    =     27,214
Dependent variable: ln_wage                       Residual df      =         11
Fixed effects: idcode year                        R-squared        =     0.6751
Std. errors: cluster (idcode ind_code)            Adj R-squared    =     0.6166
                                                  Within R-sq.     =     0.0477
                                                  Root MSE         =     0.2948
                                                  Clusters         =         12
------------------------------------------------------------------------------
     ln_wage |    Coef.     Std. err.    t     P>|t|    [95% conf. interval]
-------------+----------------------------------------------------------------
      tenure |  .0106748    .0023903    4.47   0.001     .0054138   .0159358 ***
     ttl_exp |  .0327797    .0023843   13.75   0.000     .0275319   .0380275 ***
------------------------------------------------------------------------------
* p<0.10, ** p<0.05, *** p<0.01
```

```
Singletons or perfectly predicted observations dropped: 550
Degrees of freedom absorbed by fixed effects: 4,148
```

4.5 Several outcomes at once

Put the outcomes in parentheses. R reads the data once and, when the outcomes are missing in the same observations, computes the fixed effects once, which is much faster than separate regressions. Each outcome uses its own nonmissing observations, as in separate regressions. With `estore(m)` the results are stored as `m1` and `m2`, ready for `esttab`.

```
. rfixest (ln_wage hours) tenure ttl_exp, absorb(idcode year) cluster(idcode) estore(m) ///
    quietly
Estimates stored: m1 m2

. esttab m1 m2, se star(* 0.10 ** 0.05 *** 0.01) b(%9.4f) se(%9.4f) stats(N r2_within, ///
    fmt(%9.0fc %9.4f) labels("Observations" "Within R-squared"))

--------------------------------------------
                      (1)             (2)
                  ln_wage           hours
--------------------------------------------
tenure             0.0107***       0.1465***
                  (0.0015)        (0.0329)

ttl_exp            0.0326***       0.5443***
                  (0.0024)        (0.0643)
--------------------------------------------
Observations       27,549          27,480
Within R-s~d       0.0477          0.0162
--------------------------------------------
Standard errors in parentheses
* p<0.10, ** p<0.05, *** p<0.01
```

4.6 Saving the fixed effects and the residuals

As in `reghdfe`, `absorb(`*newvar*`=`*var*`)` saves the estimated fixed effect of each observation and `residuals(`*newvar*`)` saves the residuals. Both are filled only in the estimation sample. With two or more fixed effects each one is identified only up to a constant, so the level of a saved fixed effect can differ from `reghdfe` by a constant. Differences between groups, the sum of all fixed effects plus the constant, and the residuals are the same as in `reghdfe`.

```
. rfixest ln_wage tenure ttl_exp, absorb(FE_id=idcode FE_year=year) residuals(res) ///
    cluster(idcode) quietly

. summarize FE_id FE_year res

    Variable |        Obs        Mean    Std. dev.       Min        Max
-------------+---------------------------------------------------------
       FE_id |     27,549    1.452876    .3330865   .0921975   3.414487
```

```
     FE_year |     27,549   -.0122049    .0392093   -.085001    .044967
         res |     27,549   -2.61e-17    .2715253  -1.915837   2.976263
```

4.7 Poisson regression

Use `family(poisson)` for counts or for outcomes in levels with zeros. The estimates match `ppmlhdfe`.

```
. rfixest hours tenure ttl_exp, absorb(idcode year) cluster(idcode) family(poisson)

Poisson regression with fixed effects (fixest)    Number of obs    =     27,480
Dependent variable: hours                         Residual df      =      4,141
Fixed effects: idcode year                        Pseudo R-sq.     =     0.1385
Std. errors: cluster (idcode)                     Log likelihood   =  -101128.9
                                                  Clusters         =      4,142
------------------------------------------------------------------------------
       hours |    Coef.     Std. err.    t     P>|t|    [95% conf. interval]
-------------+----------------------------------------------------------------
      tenure |  .0037454    .0008811    4.25   0.000      .002018   .0054728 ***
     ttl_exp |  .0156936    .0018564    8.45   0.000     .0120541   .0193332 ***
------------------------------------------------------------------------------
* p<0.10, ** p<0.05, *** p<0.01
Singletons or perfectly predicted observations dropped: 556
Degrees of freedom absorbed by fixed effects: 4,156
```

4.8 Postestimation

`test`, `lincom` and `e(sample)` work after `rfixest`.

```
. rfixest ln_wage tenure ttl_exp, absorb(idcode year) cluster(idcode) quietly

. test tenure = ttl_exp

 ( 1)  tenure - ttl_exp = 0

       F(  1,  4146) =   38.52
            Prob > F =    0.0000

. lincom tenure + ttl_exp

 ( 1)  tenure + ttl_exp = 0

------------------------------------------------------------------------------
     ln_wage | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
         (1) |   .0433171   .0019604    22.10   0.000     .0394737    .0471606
------------------------------------------------------------------------------

. count if e(sample)
  27,549
```

5 Speed

With three outcomes, 2 million observations and three fixed effects (100,000 individuals, 20,000 firms and 20 years), with standard errors clustered by individual, on a Windows desktop:

	Time
`reghdfe`, one call per outcome	16.3 s
`rfixest (y1 y2 y3) x1 x2, absorb(id firm year) vce(cluster id)`	3.9 s

Each call has a fixed cost of about half a second to start R, so for very small datasets `reghdfe` can be just as fast. The gains grow with the size of the data, the number of fixed effects and the number of outcomes.

6 Stored results

`e(b), e(V)`	coefficients and variance matrix
`e(N)`	observations used
`e(N_full), e(num_singletons)`	observations before and dropped as singletons
`e(N_clust)`	number of clusters (smallest dimension with multiway clustering)
`e(df_r), e(df_a)`	residual degrees of freedom and degrees of freedom absorbed by the fixed effects
`e(r2), e(r2_a), e(r2_within), e(rmse)`	fit statistics (OLS)
`e(ll), e(r2_p)`	log likelihood and pseudo R^2 (Poisson, negative binomial)
`e(depvar), e(absvars), e(clustvar)`	dependent variable, fixed effects and cluster variables
`e(cmdline)`	the command as typed
`e(sample)`	estimation sample

7 Questions and problems

"rfixest could not run R". R is not installed or is in an unusual folder. Install R, or run `rfixest setup, rscript("`*`path to Rscript`*`")`.

"The R package fixest is not installed". Run `rfixest setup`.

Does it work in batch mode? On Windows, use Stata with its normal window. On macOS and Linux it also works in batch mode.

Where are my data sent? Nowhere outside your computer. `rfixest` writes the variables of the model to a temporary file in Stata's temporary folder, R reads it, and the file is deleted at the end.

Can I use predict or margins? Not in this version. Coefficients, standard errors and all tests based on them are available.

How should I cite it? The estimation is done by `fixest`. Please cite Bergé, L. (2018), "Efficient estimation of maximum likelihood models with multiple fixed-effects: the R package FENmlm", CREA Discussion Papers 13.